SMART BUILDING CONTROLLER

Konrad Skoczylas and Javier Reyes | Year 3 Software Development, SETU Carlow | November 2026

1 What it is

A smart building controller written in C with raylib. It reads **sensors**, runs **automation rules** and operates **actuators** in a simulated six-room building (Office, Hallway, Lobby, Kitchen, Server Room and Storage). Fires, faults and intruders are the scenarios that test it, and a person can take over at any time. This document lists everything it does, then introduces the five villains.

2 Controller features

2.1 Sensors and actuators

	What the controller has
Room sensors	Temperature, motion (PIR), smoke level and smoke alarm, water leak, door state, light level
Building sensors	Battery power, sprinkler water tank, network speed, outside weather, electricity tariff
Room actuators	Lights, door lock, vent (HVAC), sprinkler valve, circuit breaker, thermostat setpoint (16 to 28 °C)
Building actuators	Main water valve, scenes, automation rules on or off

2.2 Automation rules

Seven rules run twice a second. **Local** rules always work. **Cloud** rules pause during an internet outage, and the overview then shows CLOUD OFFLINE. Each rule shows ON or OFF and how many times it has acted.

Rule	Type	Default	What it does
AUTO-FIRE	Local	On	Seals a room on a smoke or heat alarm, sprays ordinary fires, cuts power in electrical fires, releases the room when the fire is out
AUTO-VENT	Cloud	On	Opens a vent when a room passes 26 °C
AUTO-ECO	Cloud	On	Switches to ECO mode when the battery is below 25%
AUTO-LEAK	Cloud	On	Closes the main water valve when a leak is sensed
AUTO-SCHEDULE	Cloud	On	Night setback to 18 °C, comfort at 21 °C from 5 AM
AUTO-LIGHTS	Cloud	Off	Turns the lights on when motion is seen in a dark room
AUTO-DOOR	Cloud	Off	Locks the Office door while a villain is close, then unlocks it

2.3 Scenes

One-click presets that change many actuators at once. Each has a cooldown, and a scene is refused when there is no power.

Scene (key)	Effect
Lockdown (Z)	Locks every door and seals every vent for 20 seconds
Eco (X)	Lights off except the Office, thermostats to 19 °C, standby power halved
Fire Response (C)	Contains every room with fire, smoke or heat: sealed, sprayed if safe, power cut for electrical fires
Reset (V)	Back to normal: lights on, thermostats 21 °C, vents open, main valve open

2.4 Energy and comfort

- The battery drains with every light, open vent, locked door and the sprinkler pump. Locked doors cost the most.
- Night tariff of EUR 0.15 per kWh, and a morning peak of EUR 0.30 per kWh from 5 AM.

- Energy used in kWh, cost in euro and CO2 are shown live on the overview.
- Night setback (18 °C) and ECO mode reduce energy use.
- Weather outside falls from about 9 °C to 4 °C overnight. A sealed Server Room heats up to 36 °C, while sealed Storage stays near 24 °C.

2.5 Monitoring and control

Screen	Key	What it gives you
Building overview	B	Floor plan with live temperatures, status chips, meters, alarms and rule status
Control panel	SPACE	Room, Scenes, Auto, Utility (power, water, breakers) and Log tabs
Camera views	1 to 6	3D view of each room; click the door, lights, vent, sprinkler and thermostat
Graphs	TAB	Network speed, room temperatures, motion, ventilation, open versus sealed rooms
Help and pause	H, P	Help screens and hover hints; pause at any time

A smart advisor also shows one line of guidance for whatever matters most right now, for example "Grease fire in Kitchen: water makes it worse, press C to smother it".

2.6 Faults and random events

About once a minute one of these happens. Controls behave like real equipment: they refuse to work with no power, a tripped breaker, a burnt-out room or an offline door controller.

Event	What happens
Power surge	A room's breaker trips and the room goes dead until it is reset
Burst pipe	A room floods and the water tank drains until the main valve is closed
HVAC fault	For 70 seconds sealed rooms get 8 °C hotter and open vents work at half speed
Internet outage	For 40 seconds every cloud rule stops
False alarm	A smoke alarm rings for 25 seconds with no smoke
Door controller fault	A door cannot be locked or unlocked for 20 seconds

2.7 Fire and sprinklers

Fire type	Where	Water	Right response
Ordinary	Office, Hallway, Lobby, Storage	Puts it out	Seal the room and spray
Grease	Kitchen	Makes it flare up	Seal the room and let it starve
Electrical	Server Room	Shorts the circuits	Cut the room's power

- Fires grow faster as they get bigger, flash over at 85% and burn a room out after 10 seconds fully ablaze.
- Opening the door of a sealed, burning room causes a backdraft. Fire and smoke also spread through open doors and vents.
- Each sprinkler head fuses at 57 °C. All heads share one pump, so more heads means weaker spray. With no power the system runs on gravity pressure.
- A soaked room can short its breaker, and the Server Room shorts first.

2.8 How the night ends

- **You win** by reaching 6 AM. A night is 9 minutes of real time (90 seconds per simulated hour).
- **You lose** if a villain gets into the Office, fire reaches the Office, the Office stays full of smoke for 10 seconds, or the building is lost to fire.

3 The controller in action

3.1 The two main views

The building overview (B) is the controller's home screen. The camera views (1 to 6) show the same building from inside each room and use the same controls.



Building overview (B)



Camera view of the Office (1 to 6)

3.2 Reacting to problems

On the left a fire has started in the Server Room: its tile turns red, the smoke alarm is listed, and AUTO-FIRE has sealed and locked the room and counted its action (x1). On the right the internet is down: the cloud rules turn orange and the AUTOMATION box shows CLOUD OFFLINE, while the local AUTO-FIRE rule stays green.



Fire in the Server Room



Internet outage

3.3 The control panel (SPACE)

Five tabs give full manual control. The controls refuse to work, like real equipment, when there is no power, a breaker has tripped or a door controller is offline.



Room: lights, door, vent, sprinkler valve, breaker, thermostat



Scenes: four one-click presets with cooldowns



Auto: the seven rules, LOCAL or CLOUD, ON or OFF, with counters



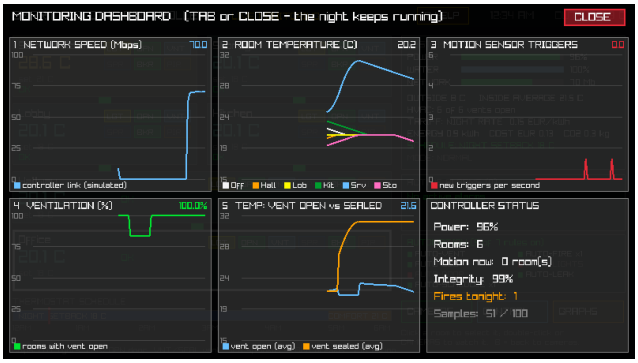
Utility: power, water, network, main valve and breakers

3.4 Log and graphs

Every action is written to the event log with its time, and five live graphs record the history of the night (TAB).



Log: every action with its time



Graphs: network, temperatures, motion, ventilation, open versus sealed

4 The villains

Five intruders wake one per hour and start far from the Office. They creep toward it along the room map, and the one who gets in ends the night with a jump scare. A locked Office door stops them: they bang on it and wander back.

Routes: Storage to Kitchen to Hallway to Office, and Server Room to Lobby to Hallway to Office.

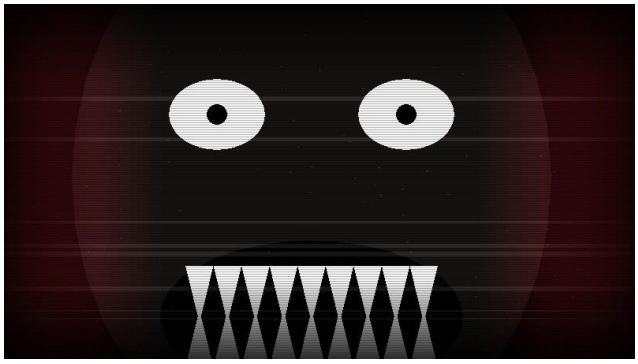
Villain	Looks like	Wakes	Starts in	Moves every	Special behaviour
Drifter	Pale humanoid	12 AM	Server Room	15 s	Walks toward the Office one room at a time
Shade	Black cloaked shape, red eyes	1 AM	Storage	12 s	Freezes while a camera is on its room
Glitch	Flickering magenta and cyan blocks	2 AM	Kitchen	18 s	Kills the lights in its room and halves the network speed
Crawler	Black spider, orange eyes	3 AM	Server Room	20 s	Uses open vents to skip a room and its locked door
Leech	Purple blob, green eyes	4 AM	Storage	22 s	Drains extra battery and slows the network

Drifter

A pale, tall figure that sways as it walks. It is the most direct threat: it simply heads for the Office. **Stop it** by locking the Office door (D) or switching on AUTO-DOOR.



Drifter on camera

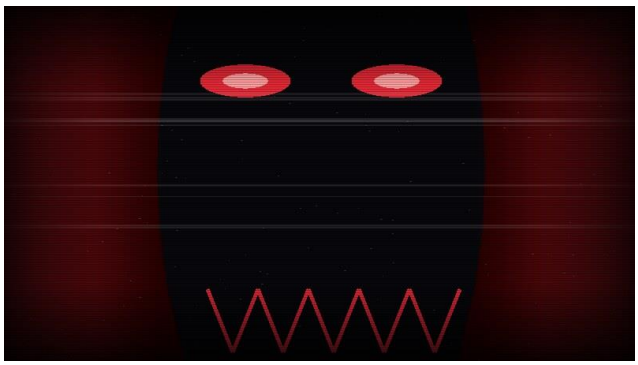

Drifter jump scare

Shade

A tall black shape with glowing red eyes that floats instead of walking. It does not move while a camera is on its room, so it only advances when you look away. **Stop it** by locking the Office door, and keep a camera on it when you can.

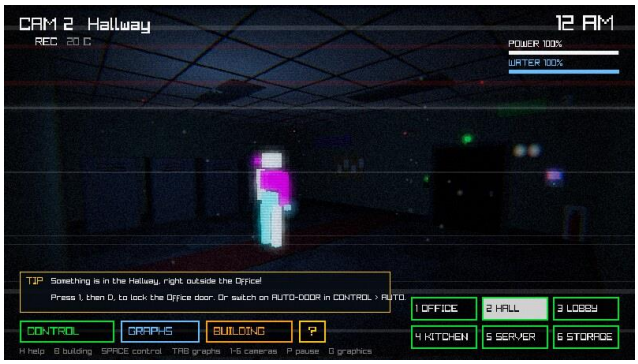


Shade on camera


Shade jump scare

Glitch

A figure built from jittering coloured blocks. It switches off the lights in whatever room it stands in and halves the network speed, which shows on the graphs. **Stop it** the same way as the others: lock the Office door.



Glitch on camera

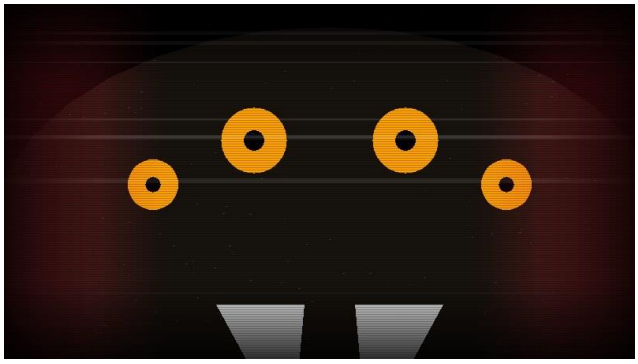

Glitch jump scare

Crawler

A six-legged spider that moves through the vents. If the next room's vent is open it can skip that room, and its locked door. **Stop it** by sealing the vents (the Lockdown scene does this) and locking the Office door.



Crawler on camera



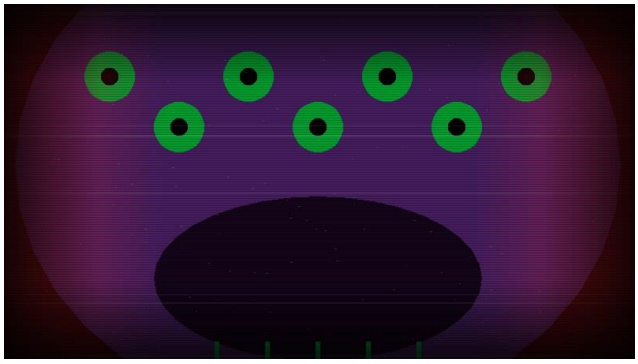
Crawler jump scare

Leech

A purple blob with several green eyes. It is slow, but while it is loose it adds a steady extra drain on the battery and cuts network speed. **Counter it** with ECO mode, and lock the Office door as normal.



Leech on camera



Leech jump scare

How the door rule helps

AUTO-DOOR locks the Office door while any villain is in the Hallway (or the Crawler is in the Lobby or Kitchen) and unlocks it when they leave. That saves power compared with keeping the door locked all night, but it is a cloud rule, so during an internet outage you must lock the door yourself.

5 Controls

Key	Action
1 to 6	Camera for Office, Hall, Lobby, Kitchen, Server, Storage
B	Building overview
SPACE	Control panel
TAB	Monitoring graphs
H, P, G	Help, pause, fancy graphics on or off
L, D, S	Lights, door lock, sprinklers of the room being viewed
Z, X, C, V	Scenes: Lockdown, Eco, Fire Response, Reset
R	Restart after the night ends